ANSWERS

Exercise 1.1

1. (i) $\begin{bmatrix} 2 & -4 \\ -6 & -3 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 1 & -1 \\ -3 & 1 & 1 \\ 9 & -5 & -1 \end{bmatrix}$ (iii) $\frac{1}{3}\begin{bmatrix} 2 & -2 & 1 \\ 2 & 1 & -2 \\ 1 & +2 & 2 \end{bmatrix}$

2. (i) $\frac{1}{2}\begin{bmatrix} -3 & -4 \\ -1 & -2 \end{bmatrix}$ (ii) $\frac{1}{28}\begin{bmatrix} 6 & -1 & -1 \\ -1 & 6 & -1 \\ -1 & -1 & 6 \end{bmatrix}$ (iii) $\frac{1}{2}\begin{bmatrix} 1 & 1 & -1 \\ -3 & 1 & 1 \\ 9 & -5 & -1 \end{bmatrix}$ 4. $A^{-1} = \frac{1}{7}\begin{bmatrix} 2 & 3 \\ -1 & -5 \end{bmatrix}$

8. $\pm\begin{bmatrix} 6 & 2 & 1 \\ 5 & 2 & 2 \\ 6 & 2 & 3 \end{bmatrix}$ 9. $\pm\frac{1}{6}\begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$ 10. $\begin{bmatrix} 2 & 0 & -2 \\ 0 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix}$

12. $\begin{bmatrix} 3 & 1 \\ 1 & -2 \end{bmatrix}$ 13. $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ 15. HELP

Exercise 1.2

1. (i) 1 (ii) 2 (iii) 2 (iv) 3 (v) 3 2. (i) 2 (ii) 3 (iii) 3

3. (i) $\begin{bmatrix} -2 & 1 \\ -5 & 2 \end{bmatrix}$ (ii) $\begin{bmatrix} -2 & -3 & 1 \\ -3 & -3 & 1 \\ -2 & -4 & 1 \end{bmatrix}$ (iii) $\begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$

Exercise 1.3

1. (i) $x = -11, y = 4$ (ii) $x = 2, y = -4$
   (iii) $x = 2, y = 3, z = 4$ (iv) $x = 3, y = -2, z = 1$
2. $x = 2, y = 1, z = -1$ 3. ₹ 18000, ₹ 600
4. 18 days, 36 days 5. ₹ 2000, ₹ 1000, ₹ 3000

Exercise 1.4

1. (i) $x = -2, y = 3$ (ii) $x = \frac{1}{2}, y = 3$
   (iii) $x = 2, y = 3, z = 4$ (iv) $x = 1, y = 3, z = 3$
2. 84 3. 50% acid is 6 litres, 25% acid is 4 litres
4. Pump A : 15 minutes, Pump B : 30 minutes
5. ₹ 30/-, ₹ 10/-, ₹ 30/-, yes

Exercise 1.5

1. (i) $x = -1, y = 4, z = 4$ (ii) $x = 3, y = 1, z = 2$
2. $a = 2, b = 1, c = 6$ 3. ₹ 30000, ₹ 15000, ₹ 20000
4. $a = 1, b = 3, c = -10$, yes

Exercise 1.6

1. (i) $x = y = z = 1$ (ii) $x = \frac{1}{10}(7 - 5t), y = \frac{1}{10}(5t - 1), z = t, t \in \mathbb{R}$

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